MOSAM BISWAS

Software Engineer · AI/ML · Full-Stack · Dombivli (East), Maharashtra, India

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SUMMARY

B.Tech AI & Data Science (2026) graduate who *builds AI systems and ships them as tested software*: a **fully on-device voice assistant**, an **LLM-agent platform** that usability-tests apps, an **AI form-filling Chrome extension**, and offline apps for two real businesses. **IEEE-published**; immediate joiner, open to relocation.

TECHNICAL SKILLS

LANGUAGES	Python, JavaScript, TypeScript, Java, Dart, Go, SQL, C++, C, HTML/CSS
AI/ML & LLMS	PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, SHAP, OpenCV, GANs, LLM Agents, Ollama, OpenAI / Gemini / Claude APIs, Whisper, Local LLM Inference, Structured Outputs (JSON Schema)
BACKEND & WEB	FastAPI, Flask, Node.js, Express.js, Next.js, React, Angular, REST APIs, WebSockets, Chrome Extensions
MOBILE & DATA	Flutter, Android (Java), SQLite, MySQL, PostgreSQL, MongoDB, Firebase, Prisma, Pandas, Streamlit
DEVOPS & PRACTICES	Git, GitHub Actions (CI/CD), Linux, AWS, Unit & E2E Testing, Postman, Agile/Scrum

EXPERIENCE

SOFTWARE DEVELOPER INTERN at *Sedna Technologies*

Jan 2026 – Apr 2026 · Mumbai, India

- Shipped work across **4 client projects**, working directly with clients to gather requirements and resolve issues.
- Designed and built a **client-facing dashboard solo** with **Angular, Flask, and MySQL over REST APIs**, owning it from requirements to handover; later submitted as my B.Tech dissertation.

PROJECTS

AMINAL

Jul 2026 – Present

Python, Whisper, Ollama (Llama 3.2, Qwen2.5-VL), Piper TTS, OpenCV, SQLite

- Built a **local-first voice assistant for Windows** that runs speech recognition, LLM intent parsing, vision, and neural text-to-speech **entirely on-device** on an RTX 4060: **29K lines of Python** across 72 modules, **125 voice commands**, zero cloud services in the voice path.
- Added code-level guards after the LLM **hallucinated completed actions** despite prompt rules; drives apps by control name via **Windows UI Automation**; backed by **849 regression tests**, each from a real failure.

LOCALS

Nov 2026 – Present

Python, LLM Agents, Ollama (Qwen3), Chrome Automation, Android Emulator

- Built an **AI usability-testing platform** where **114 simulated users from 9 job families** use web apps in a real browser and Android apps on an emulator, flag where they got stuck, and replay every step with screenshots on a live dashboard.
- Enforced human limits (patience, reading, touch-slip rates) **in code from published studies**, not the prompt; a two-stage notice-then-act agent design reached **83% task success against a 78% real-user benchmark** on a 4B local model.

FILL.AI

Sep 2026

JavaScript, Chrome Extension (MV3), Ollama, Gemini API, Claude API, pdf.js, GitHub Actions

- Built a Chrome extension that turns a resume PDF into a structured profile and **auto-fills job application forms** (Greenhouse, Lever, Google Forms, React widgets, iframes) with a local or cloud LLM behind **one JSON schema**.
- Blocked hallucinated answers in code: every answer must **cite a profile field**, contact details and dates are verified, and each fill is read back; covered by **121 end-to-end checks** with **CI/CD releases** on GitHub Actions.

PROJECTS

MINDFIRST
Apr 2025

Python, Flask, Next.js, Prisma, NextAuth, Tailwind CSS

- Developed a **mental health pre-screening platform used by 100+ students**, combining a chatbot with PHQ-9/GAD-7 psychometric analysis at **95% accuracy**.
- Integrated NextAuth authentication and personalized reports, **reducing user drop-off by 30%**.

POPS & ORDER CONTROL
2026

Flutter, Dart, SQLite, Android

- Built **two offline Android apps for real small businesses**: an order tracker that replaced a fabrication firm's paper diary, and a food stall order board on transactional SQLite with 13 unit tests.

SHELLFOLIO
Sep 2026

Go, SSH (Wish), Bubble Tea, Linux, systemd, Let's Encrypt

- Deployed a terminal portfolio (`ssh mosambiswas.com`) as **one Go binary on a Linux VPS** with auto-renewed TLS, **per-IP rate limiting**, session caps, and a least-privilege systemd service.

PIXELFORGE
2025

Python, PyTorch, GANs, Computer Vision

- Classified leukemia cells on the C-NMC dataset at **0.90 ROC-AUC and 95% recall**, using a custom spectral-norm GAN to augment scarce training data.

RESEARCH & PUBLICATIONS

Comparative Analysis of Psychometric Data for Assessing Mental Health Testing Needs Using Machine Learning Models

Co-author · IEEE Xplore · Proc. ICAIIHI 2025 · DOI: 10.1109/ICAIIHI67124.2025.11403838

- Benchmarked Logistic Regression, Random Forest, SVM, XGBoost, and LightGBM on **28.5K PHQ-9/GAD-7 records** (stratified 5-fold CV, paired t-tests); Logistic Regression led at **99% accuracy** with SHAP explanations.

EDUCATION

RAMRAO ADIK INSTITUTE OF TECHNOLOGY · *D. Y. Patil Deemed to be University*
2022 – 2026

B.Tech in Computer Engineering (Major: Artificial Intelligence and Data Science) · **CGPA: 7.5** · Navi Mumbai, India

CERTIFICATIONS, LEADERSHIP & COMMUNITY

CERTIFICATIONS	Samsung Innovation Campus: AI · AWS Academy Graduate: Machine Learning · Generative AI (edX) · Agile & Scrum Fundamentals · Android Compose Camp (GDSC) · Industry-Focused Java Workshop (CSI)
LEADERSHIP	**Technical Head, Photocircle RAIT** (2024 – 2025): ran 3 workshops (75+ students), **raising participation 60%**
COMMUNITY	Microsoft Learn Student Ambassador · Google Developer Groups (25+ codelabs, Google Cloud Innovator) · Hacktoberfest contributor · Smart India Hackathon 2025 (internal round)

LET'S TALK

Open to full-time roles · immediate joiner · open to relocation

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